

Madhav Tyagi

B.Tech in Computer Science and Engineering | Bennett University, Greater Noida
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Summary: Software Engineering student with strong foundations in **data structures, algorithms, operating systems, and computer networks**. Experienced in building scalable backend systems, real-time AI solutions, and applied machine learning. Proven problem-solver with multiple national and international hackathon wins, passionate about building intelligent, reliable, and impact-driven software systems.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/%	Year
B.Tech CSE	Bennett University, Greater Noida	9.84	2023–2027
Senior Secondary	Pragyan School, Greater Noida	95%	2022
Secondary	Pragyan School, Greater Noida	96.6%	2020

EXPERIENCE

- Project Intern – RAC, DRDO**

Generative AI, LLMs, Python, LangChain

Hybrid

June 2025 – December 2025

 - Designed and deployed end-to-end **LLM-driven workflows** for automated candidate evaluation and context-aware response generation.
 - Built modular, scalable backend services using Python to integrate LLM pipelines with structured data sources.
 - Implemented secure data ingestion, preprocessing, and validation pipelines to ensure reliability in sensitive workflows.
 - Optimized API performance and conducted systematic testing to improve system **scalability, robustness, and fault tolerance**.

PROJECTS

- PRAVAH – Intelligent Traffic Optimization System (SIH 2025 Winner)**

Reinforcement Learning, YOLO, SUMO, FastAPI, React

GitHub

 - Built a **3-layer AI-driven traffic optimization system** to replace fixed-timer signals.
 - Optimization:** Real-time traffic density detection using CCTV feeds and YOLO-based computer vision.
 - Prediction:** Reinforcement learning models trained on historical data to proactively prevent congestion.
 - Prevention:** Commuter-facing system suggesting optimal departure times to avoid peak traffic.
- MIRA 2.0 – AI-Powered Recruitment Platform (SIH 2024 Winner)**

Python, Streamlit, OLLAMA, LLaMA, ETL, LLMs

GitHub

 - Automated resume parsing, candidate shortlisting, and expert interviewer mapping using LLM pipelines.
 - Designed ETL workflows with semantic scoring for high-accuracy profile–role matching.
 - Deployed scalable backend APIs with reliability-focused testing and debugging.
- SAHARA – Elderly Care Companion**

YOLOv8, Whisper, Gemini, Flask, SQLite

GitHub

 - Developed an AI assistant for senior citizens with voice reminders, fall detection, and emergency alerts.
 - Integrated secure authentication, Google Calendar sync, and real-time monitoring.
- God’s Eye – Surveillance Detection Platform**

YOLOv5, FaceNet, FastAPI, Next.js, Firebase

GitHub

 - Built a full-stack system for crowd-based person detection and license plate recognition.
 - Implemented low-latency APIs, authentication, and secure backend infrastructure.

SKILLS

- Programming Languages:** Python, C++, C, Java, JavaScript, HTML, CSS, R
- Core CS:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP
- Backend & Systems:** FastAPI, Flask, REST APIs, Authentication, ETL Pipelines, SQL/NoSQL Databases
- AI / ML:** Reinforcement Learning, Computer Vision (YOLO), Deep Learning (CNNs), LLMs, Prompt Engineering
- Tools & DevOps:** Git, Docker, Firebase, Streamlit, SUMO, Linux, Cloud Deployment (AWS/GCP)

ACHIEVEMENTS & RESPONSIBILITIES

- Winner – Smart India Hackathon**

2025
- Winner – Cyber AI Hackathon, University of Derby**

2025
- Winner – Agentic AI Hackathon, Ulster University (UK)**

2025
- Winner – Smart India Hackathon**

2024
- Top 5 Finalists – Microsoft Innovate Hackathon**

Jan 2025